In [ ]:
```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# STEP 1 & 2: Load Image & Convert to Grayscale    // Pragya Singhal,CU24220050,43
# Load an image (replace 'path_to_image.jpg' with your image path or use a sample i
# If no local image is available, we can create/load a dummy low-contrast grayscale
image_path = "img.jpg.jpg"  # TODO: Update with actual image path
original_img = cv2.imread(image_path)

if original_img is None:
  # Fallback: Create a synthetic low-contrast image for demonstration if file not f
  print(
      "[INFO] Image not found at path. Generating a synthetic low-contrast"
      " test image..."
  )
  x = np.linspace(0, 255, 300)
  y = np.linspace(0, 255, 300)
  xx, yy = np.meshgrid(x, y)
  original_img = np.clip(
      np.sin(xx / 30) * 30 + 100, 50, 150
  ).astype(  # Low contrast range [50, 150]
      np.uint8
  )
  gray_img = original_img
else:
  # Convert color image to grayscale
  if len(original_img.shape) == 3:
    gray_img = cv2.cvtColor(original_img, cv2.COLOR_BGR2GRAY)
  else:
    gray_img = original_img


# STEP 4: Contrast Stretching (Min-Max Normalization)      // Pragya Singhal,CU2422

def contrast_stretching(img):
  r_min, r_max = np.min(img), np.max(img)
  # Prevent division by zero if image is uniform
  if r_max == r_min:
    return img
  # Stretch pixel intensities to the full [0, 255] range
  stretched = ((img.astype(np.float32) - r_min) / (r_max - r_min)) * 255.0
  return np.clip(stretched, 0, 255).astype(np.uint8)


stretched_img = contrast_stretching(gray_img)


# STEP 5: Histogram Equalization (Global)

equalized_img = cv2.equalizeHist(gray_img)

# STEP 6: Contrast Limited Adaptive Histogram Equalization (CLAHE)
# clipLimit controls the contrast limit, tileGridSize defines the grid dimensions
```

```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
clahe_img = clahe.apply(gray_img)

# STEP 3 & 7: Plotting Images and Histograms
fig, axes = plt.subplots(4, 2, figsize=(12, 16))

images = [gray_img, stretched_img, equalized_img, clahe_img]
titles = [
    "Original Low-Contrast Image",
    "Contrast Stretched Image",
    "Histogram Equalized Image",
    "CLAHE Enhanced Image",
]

for i in range(4):
  # Row i, Column 0: Image
  axes[i, 0].imshow(images[i], cmap="gray", vmin=0, vmax=255)
  axes[i, 0].set_title(titles[i])
  axes[i, 0].axis("off")

  # Row i, Column 1: Histogram
  axes[i, 1].hist(
      images[i].ravel(), bins=256, range=[0, 256], color="black", alpha=0.7
  )
  axes[i, 1].set_title(f"Histogram of {titles[i]}")
  axes[i, 1].set_xlim([0, 256])
  axes[i, 1].set_xlabel("Pixel Intensity")
  axes[i, 1].set_ylabel("Frequency")

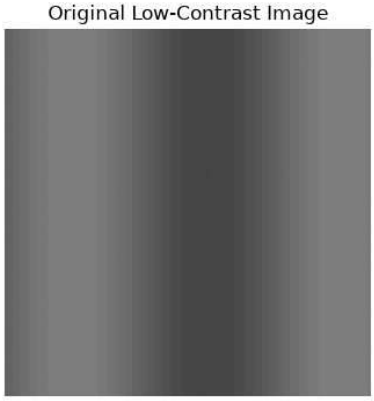
plt.tight_layout()
plt.show()


# STEP 8, 9 & 10: Summary & Analysis
print("\n" + "=" * 60)
print("EXPERIMENT OBSERVATIONS & DISCUSSION")
print("=" * 60)
print(
    "1. Original Image: Displays a restricted dynamic range with compressed"
    " pixel intensities, leading to poor visibility and low contrast."
)
print(
    "2. Contrast Stretching: Stretches the minimum and maximum intensity"
    " values to cover the full [0, 255] range, effectively improving global"
    " contrast."
)
print(
    "3. Histogram Equalization: Flattens the histogram globally by redistributing"
    " pixel intensities based on their cumulative distribution function (CDF)."
    " While effective for uniform illumination, it can over-enhance noise in"
    " smooth regions."
)
print(
    "4. CLAHE: Operates on small local regions (tiles) and clips the histogram"
    " to prevent over-amplification of noise. It outperforms global"
    " equalization on images with uneven lighting (e.g., shadows or varying"
```

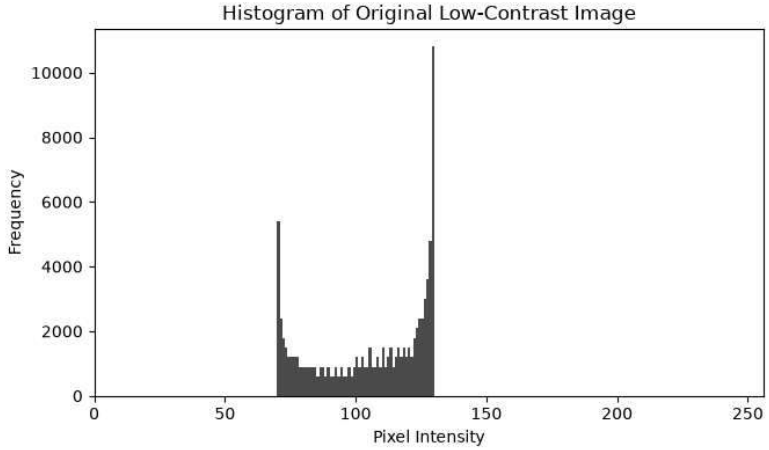
```
    " illumination)."
)
```

```
[INFO] Image not found at path. Generating a synthetic low-contrast test image...
[ WARN:0@2.140] global loadsave.cpp:278 findDecoder imread_('img.jpg.jpg'): can't op
en/read file: check file path/integrity
```

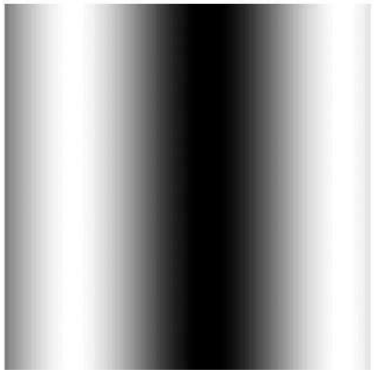
Original Low-Contrast Image



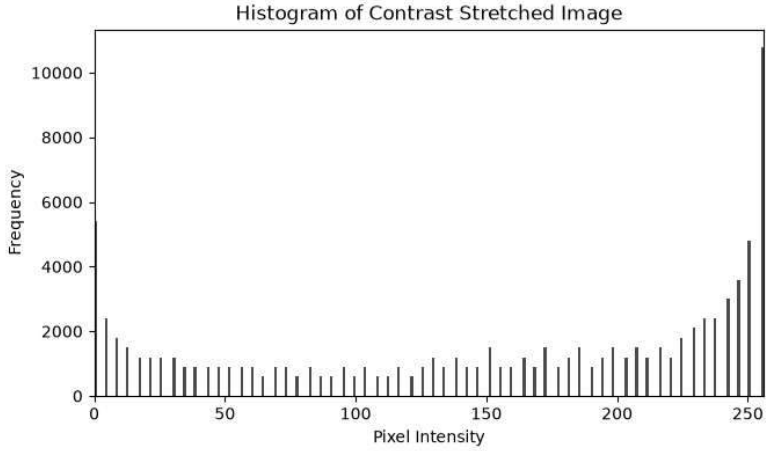
Histogram of Original Low-Contrast Image



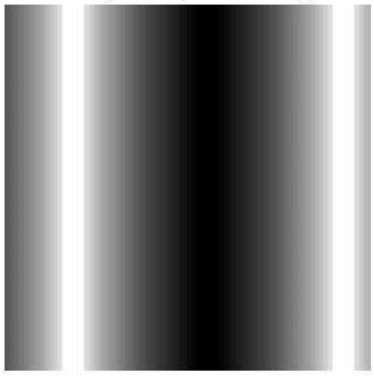
Contrast Stretched Image



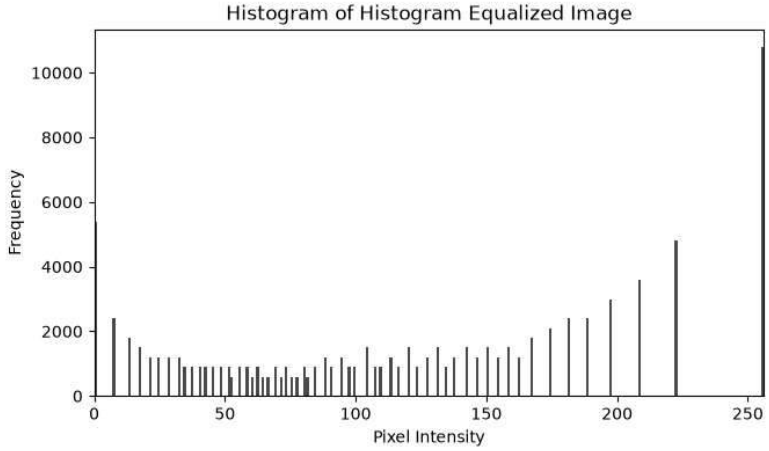
Histogram of Contrast Stretched Image



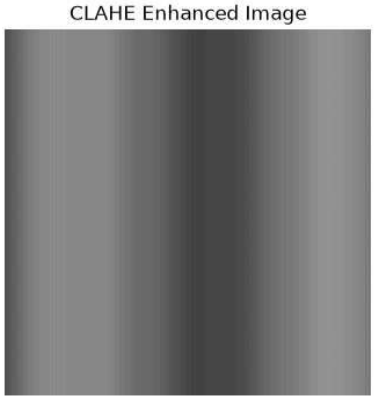
Histogram Equalized Image



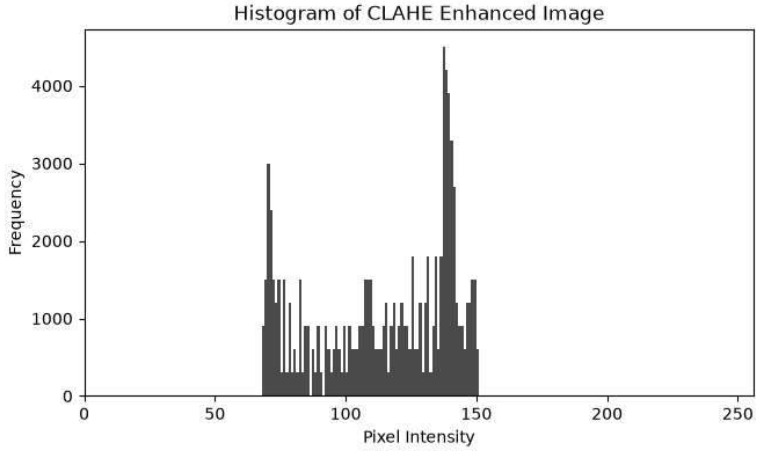
Histogram of Histogram Equalized Image



CLAHE Enhanced Image



Histogram of CLAHE Enhanced Image



```
============================================================
EXPERIMENT OBSERVATIONS & DISCUSSION
============================================================
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